

Executive Summary:

Equation Group is a highly sophisticated cyber threat actor, widely believed to be linked to the U.S. National Security Agency (NSA). It is considered one of the most advanced hacking groups in history, known for developing and deploying powerful cyber-espionage tools. The group is strongly associated with the NSA's Tailored Access Operations (TAO) unit and has been active since at least the 1990s, with major operations identified between 2001 and 2015. Equation Group primarily targets governments, military organizations, financial institutions, telecommunications networks, and the energy and aerospace sectors.

Its malware arsenal includes highly sophisticated tools such as *EquationDrug*, *DoubleFantasy*, *GrayFish*, *Fanny*, and *Equestre*, many of which exhibit unparalleled persistence and stealth capabilities. One of its most notable tactics involves infecting hard drive firmware, making the malware nearly impossible to remove. Additionally, Equation Group has developed a range of zero-day exploits, some of which were leaked in 2016 by *The Shadow Brokers*, a mysterious hacking group. These leaks, including the infamous *EternalBlue* exploit, later fueled global cyberattacks like WannaCry. Despite exposure, Equation Group remains a benchmark for cyber warfare, influencing modern Advanced Persistent Threats (APTs) and shaping the evolution of state-sponsored cyber operations worldwide.

Here we are talking about a particular malware which was developed by Equation Group known as GrayFish, it was a Win32 EXE file of 560.00 KB which was first seen in 2022-02-08 .

Its File hash is:

SHA-1 5283da3297d7c7cce1d911f5f411206c8734c18a

SHA-256 248c45236a4d4b5aec67b01a1e2e42a15bb6d2de0dee296f345d976ac01c3646

And it was related to Domain:

arc.msn.com

It is the subdomain of

msn.com:

Its Siblings are:

betagames.msn.com

cimail200.msn.com

dnekan.msn.com

fults.msn.com

htmllatino.msn.com

lenovo15.msn.com

msnbot-65-55-208-39.msn.com

screeningroom.msn.com

th.msn.com

yildirayyilmaz.msn.com

The particular NS Records of this domain is:

- DNS1.P09.NSONE.NET
- NS1-204.AZURE-DNS.COM
- NS2-204.AZURE-DNS.NET
- NS3-204.AZURE-DNS.ORG
- NS4-204.AZURE-DNS.INFO

And the Possible C2s is:

20.99.185.48

And Connected Ips are:

- 13.107.39.203
- 20.82.210.154
- 95.101.28.33
- 95.101.28.49

Methodology

Tools used For finding siblings:

[Welcome to Robtex!](#)

Tools used For finding NS Records:

DNSDumpster

Whois Lookup

Tools used For information and logs gathering:

Virustotal

OSINT platforms:

WHOIS records

[Equation Group - Wikipedia](#)

[0x00sec - The Home of the Hacker](#)

[Throwback attack: The Equation Group, 'God of cyberespionage' - Control Engineering](#)

[A Deep Dive into DoubleFeature, Equation Group's Post-Exploitation Dashboard - Check Point Research](#)

[Equation Group: The Crown Creator of Cyber-Espionage](#)

Detailed Analysis

Although the domain it self doesn't look suspicious but is communicated and referred many malicious files some of them are:

Communicating Files:

- 000005427fbe8d0b914fc56eec86ca6ee480a3a44b5fb5cb19eaeec29c21240e
- Bfa700929ef518911fe10a449b4b103c
- 597c36314d96c24ce622f232404fb6d0
- 00000be9b714ac45f7607a9c4d28b6be625f83015ba2e1c4a3ef234cc126f7ff
- 00000a931ada172ac7cb8acdf3ae89046ec1ac7b46a3e5edb7861bb2bfbd9562
- q2g54WndLShXYB3BIA5J

Files Referred:

- 04fc01e3bfa4aa8893167fd814a356d1cdea601b82c866f2e4ccc032b22e3d78
- 4e2391c0c7b4e0ce5941f2574b6d882263452fc5c5adcbbb3cef35e02a1a686c
- 6850b3660aaaadb021045af0d0a165d22ee0347fb15ee54c770fcf0e82e3d61e
- 7323aa9bb112059dba086f8b2a49afcc8eeff4ebfd3dffa6cdc1c8ec8fdb45cd
- 542fcd0a40d24bc242dab43651634f3b3ee86ce6d09d6f3318a672572eca9a79

Discovered forum or onion site references to Equation Group

- <http://w27irt6ldaydjoacyovepuzlethuoy pazhhbot6tljuywy52emetn7qd.onion/podcasts/Security%20Now/sn/Security%20Now%20574%20show%20transcript.htmlv>
- <http://w27irt6ldaydjoacyovepuzlethuoy pazhhbot6tljuywy52emetn7qd.onion/cons/SteelCon/SteelCon%202019/How%20The%20Equation%20Group%20Remained%20Out%20Of%20Sight%20Over%20The%20Years%20-%20Connor%20Morley.mp4>

Evidence of malicious behavior or C2 hosting.

A user discovered that his system is getting connected to arc.msn.com multiple time without making a connection request.

Source: [Who is my PC talking to?. Windows periodically calls into various... | by David Lee | Medium](#)

- SystemSettings connected 4 times to arc.msn.com. Again, I have no idea why.
- TextInputHost connected once to arc.msn.com for unknown reasons.

One similar case:

Source: [Do not connect external server - User Group - Spiceworks Community](#)

5. These external servers are automatically connected. In my case, about 7 MB was automatically downloaded when I first connected to the Internet.

- [dns.msftncsi.com](#) 1 
- [bing.com](#) 
- [wdcp.microsoft.com](#) 1 
- [licensing.md.mp.microsoft.com](#) 
- [ocsp.digicert.com](#) 
- [wdcpalt.microsoft.com](#) 
- [checkappexec.microsoft.com](#) 2 
- [arc.msn.com](#) 
- [client.wns.windows.com](#) 

Attribution & TTPs

Execution(TA0002)

Scripting(T1064)

Privilege Escalation(TA0004)

Process Injection(T1055)

Defense Evasion(TA0005)

Obfuscated Files or Information(T1027)

Software Packing(T1027.002)

Process Injection(T1055)

Scripting(T1064)

Discovery(TA0007)

System Information Discovery (T1082)

Conclusion & Recommendations

The Equation Group is a highly advanced APT known for sophisticated cyber-espionage. Their tools and techniques demand proactive cybersecurity measures to mitigate threats effectively.

Mitigation Strategies

Domain & IOC Blocking – Block known malicious domains, monitor for new IOCs, and analyze network traffic.

Network & Endpoint Security – Implement segmentation, DNS filtering, EDR solutions, and regular patching.

Access Control & Awareness – Enforce MFA, least privilege access, and train personnel on phishing risks.

Future Intelligence Collection

OSINT & Dark Web Monitoring – Track threat actor discussions and leaked tools.

Infrastructure Analysis – Identify new C2 domains using passive DNS and WHOIS lookups.

Malware Research – Reverse-engineer new samples to detect evolving tactics.

TTP Evolution Tracking – Monitor changes in MITRE ATT&CK techniques and industry threat reports.

Appendices:

248c45236a4d4b5aec67b01a1e2e42a15bb6d2de0dee296f345d976ac01c3646

43

/ 68

Community Score

43/68 security vendors flagged this file as malicious

Reanalyze

Similar

248c45236a4d4b5aec67b01a1e2e42a15bb6d2de0dee296f345d976ac01c3646

DOGROUND.exe

peexe

Size

560.00 KB

Last Analysis Date

3 years ago

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 6

Crowdsourced YARA rules ⓘ

⚠

Matches rule EquationGroup_EquationDrug_Gen_3 from ruleset apt_eqgrp at https://github.com/Neo23x0/signature-base by Auto Generated

↳ EquationGroup Malware - file mssld.dll

⚠

Matches rule EquationGroup_Toolset_Apr17_DiBa_Target_2000 from ruleset apt_eqgrp_apr17 at https://github.com/Neo23x0/signature-base by Florian Roth

↳ Detects EquationGroup Tool - April Leak

⚠

Matches rule EquationGroup_Toolset_Apr17_DoubleFeatureDll_dll_3 from ruleset apt_eqgrp_apr17 at https://github.com/Neo23x0/signature-base by Florian Roth

↳ Detects EquationGroup Tool - April Leak

⚠

Matches rule apt_equation_exploitlib_mutexes from ruleset spy_equation_fiveeyes at https://github.com/Neo23x0/signature-base

↳ Rule to detect Equation group's Exploitation library http://goo.gl/ivt8EW



arc.msn.com



Communicating Files (1 M) ⓘ

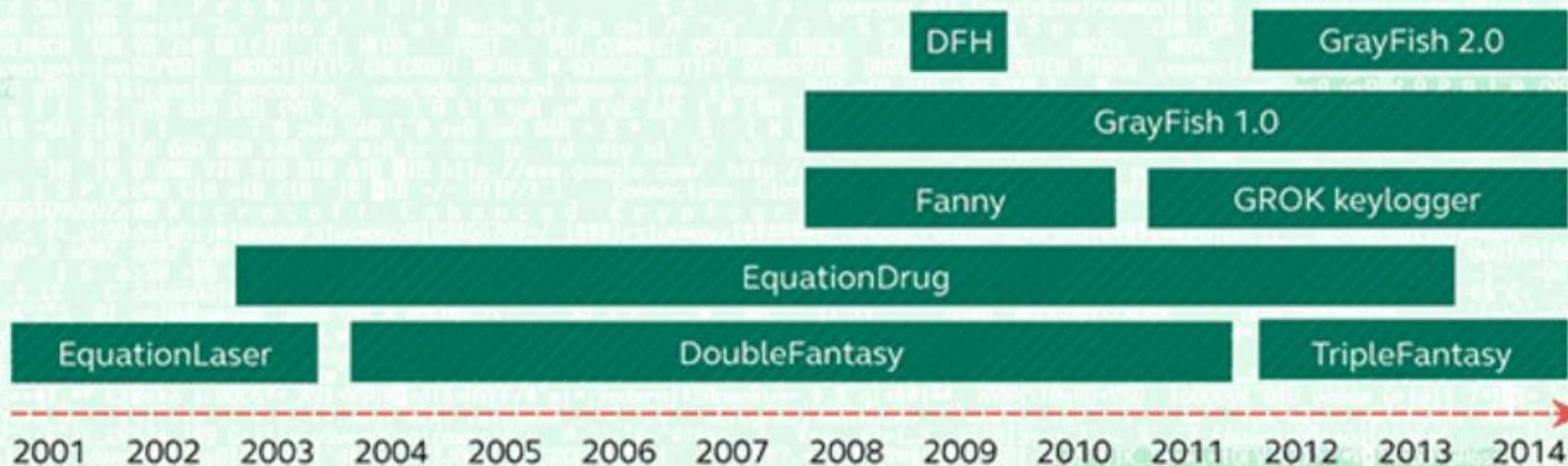
Scanned	Detections	Type	Name
2025-03-03	67 / 72	Win32 EXE	parent.txt
2025-03-05	62 / 72	Win32 EXE	snmptrap.exe
2024-04-05	67 / 72	Win32 EXE	000002305f386d9f7223c3bbf47164ca6f09f947dcd83b54c657594c54c6a359.exe
2025-03-13	71 / 73	Win32 EXE	EXPLORER.SCR
2025-02-14	7 / 72	Win32 EXE	Au_.exe
2025-03-02	68 / 72	Win32 EXE	1.exe
2024-01-24	1 / 70	Win32 EXE	Warcraft III Launcher
2024-01-24	54 / 69	Win32 EXE	ODSERV
2024-12-09	64 / 71	Win32 EXE	conwurm.exe
2024-01-24	66 / 70	Win32 EXE	18jm2a4.exe

...

Files Referring (200) ⓘ

Scanned	Detections	Type	Name
2025-03-13	20 / 73	Win32 EXE	walliant.exe
2025-03-13	28 / 72	Win32 EXE	Barousel.exe
2025-03-13	30 / 73	Win32 EXE	Barousel.exe
2025-03-13	19 / 73	Win32 EXE	Walliant.exe
2025-03-13	22 / 73	Win32 EXE	Walliant.exe
2025-03-13	29 / 73	Win32 EXE	Walliant.exe
2025-03-13	20 / 73	Win32 EXE	Barousel.exe

Equation group's malware timeline



- The Fanny timeline is based on C&C server IP activity
- TripleFantasy compilation timestamps seem to be fake, graph based on compiler versions and C&C registrations
- EquationDrug appears to have been replaced with Grayfish in somewhere in 2013; development started earlier
- DFH refers to "Disk from Houston"

SHARED



This section shows related hostnames and ipnumbers

Using as CNAME

[dynmsg.modpim.com](#)

1 results shown.

Siblings

Siblings are domains or hostnames on the same level, under the same parent level. Not necessarily related in any other way

[betagames.msn.com](#)

[cimail200.msn.com](#)

[dnekan.msn.com](#)

[fults.msn.com](#)

[htmllatino.msn.com](#)

[lenovo15.msn.com](#)

[msnbot-65-55-208-39.msn.com](#)

[screeningroom.msn.com](#)

[th.msn.com](#)

[yildirayyilmaz.msn.com](#)

Equation group malware family

